

Product Expansion on Coprime Subgroup Grids

Theorem. Fix $d \geq 2$, $0 < \varepsilon < 1$, and $K \geq 1$, and write $[d] = \{1, \dots, d\}$. There is a constant $\rho = \rho(d, \varepsilon, K) > 0$ such that the following holds. Let q be prime. For $S \subseteq \mathbb{F}_q$, write $\text{RS}(S, k) = \{(f(\alpha))_{\alpha \in S} : \deg f < k\}$. Let $S_i \leq \mathbb{F}_q^\times$ have pairwise coprime orders $n_i = |S_i|$ satisfying $K^{-1} \leq n_i/n_j \leq K$ for all $i, j \in [d]$.

Put $\Omega = \prod_{i=1}^d S_i$ and identify $\mathbb{F}_q^\Omega$ with $\bigotimes_{i=1}^d \mathbb{F}_q^{S_i}$. A direction- i line is obtained by fixing every coordinate except the i -th; it is full if its i -th coordinate ranges over all of S_i . Let $1 \leq k_i \leq (1 - \varepsilon)n_i$, and set $C_i = L_i \text{RS}(S_i, k_i)$, where each $L_i: \mathbb{F}_q^{S_i} \rightarrow \mathbb{F}_q^{S_i}$ is invertible diagonal. Put

$$Q_i = \mathbb{F}_q^{S_1} \otimes \dots \otimes C_i \otimes \dots \otimes \mathbb{F}_q^{S_d}.$$

For $M \in \mathbb{F}_q^\Omega$, let $|M|$ denote its Hamming weight; for $A_i \in Q_i$, let $|A_i|_i$ denote the number of direction- i lines on which A_i is nonzero. Then every $M \in Q_1 + \dots + Q_d$ admits a decomposition $M = A_1 + \dots + A_d$, with $A_i \in Q_i$, such that

$$|M| \geq \rho \sum_{i=1}^d n_i |A_i|_i.$$

There is a universal constant $C_0 \geq 1$ for which one may take

$$\rho = \frac{1}{2dK^d} \left(\frac{\varepsilon}{(C_0 d K)^{10d}} \right)^{3d6^{d-2}}.$$

Proof.

1. Dual normalization

Let $S \leq \mathbb{F}_q^\times$ have order n , and set $D_S = \text{diag}(\alpha)_{\alpha \in S}$. If $\deg f < k$ and $\deg g < n - k$, every monomial occurring in $\alpha f(\alpha)g(\alpha)$ has exponent between 1 and $n - 1$. Therefore $\sum_{\alpha \in S} \alpha f(\alpha)g(\alpha) = 0$. A dimension comparison gives

$$D_S \text{RS}(S, k) = \text{RS}(S, n - k)^\perp. \quad (1)$$

For each i , set $r_i = n_i - k_i \geq \varepsilon n_i$ and $R_i = \text{RS}(S_i, r_i)$. Apply the global diagonal operator $D = \bigotimes_{i=1}^d (D_{S_i} L_i^{-1})$. It preserves point support and active-line sets and, by (1), sends Q_i to the corresponding space with $C_i = R_i^\perp$. It is therefore enough to prove the theorem under this normalization.

Put $m = \min_i n_i$, $N = |\Omega|$, and $\mathbf{r} = (r_1, \dots, r_d)$. Let

$$\mathcal{B}_{\mathbf{r}} = \{P \in \mathbb{F}_q[X_1, \dots, X_d] : \deg_{X_i} P < r_i \text{ for all } i\}.$$

For $U \subseteq \Omega$ and a polynomial space $\mathcal{V}$, let $\text{ev}_U(\mathcal{V})$ denote its evaluation on U . Tensor-product duality gives

$$(Q_1 + \cdots + Q_d)^\perp = R_1 \otimes \cdots \otimes R_d = \text{ev}_\Omega(\mathcal{B}_r). \quad (2)$$

2. Fixed-box closure

For $0 \leq t < \min_i n_i$, define $\mathcal{P}_t = \{P \in \mathbb{F}_q[X_1, \dots, X_d] : \deg_{X_i} P \leq t \text{ for all } i\}$. For $A \subseteq \Omega$, define its locator space

$$I_t(A) = \{P \in \mathcal{P}_t \mid P|_A = 0\}. \quad (3)$$

For $W \subseteq \Omega$, define

$$\text{cl}_t(W) = \{x \in \Omega \mid P|_W = 0 \implies P(x) = 0 \text{ for every } P \in \mathcal{P}_t\}.$$

This is the linear-matroid closure of the evaluation functionals, hence is idempotent. A set T is called t -closed if $T = \text{cl}_t(T)$. We use the same notation for closure inside any coordinate subproduct, with the corresponding polynomial variables. All evaluation matrices are defined over $\mathbb{F}_q$, so scalar extension to $\overline{\mathbb{F}_q}$ does not alter this closure.

Lemma 1 (Closure size). *If $T = \text{cl}_t(W)$ and $u = t + 1$, then*

$$|T| \leq \frac{N}{u^d} |W|. \quad (4)$$

Proof. Choose independently and uniformly subsets $J_i \subseteq S_i, |J_i| = u$, and put $J = \prod_i J_i$.

Tensor-product Vandermonde interpolation gives an isomorphism $\mathcal{P}_t \longrightarrow \mathbb{F}_q^J$. Thus the evaluation functionals at the points of $T \cap J$ are independent. On the other hand, every evaluation functional at a point of T lies in the span of those at W . Hence $|T \cap J| \leq |W|$. Taking expectations gives $|T| \frac{u^d}{N} \leq |W|$. This proves (4). $\square$

Lemma 2 (Slices). *Suppose $A \subseteq \prod_{i=1}^e S_i$ is t -closed. For $x = (x_1, \dots, x_e)$, let $x_{\hat{i}}$ be the tuple obtained by deleting its i -th coordinate. For $\alpha \in S_i$, let $A_{i,\alpha} = \{x_{\hat{i}} : (x_1, \dots, \alpha, \dots, x_e) \in A\}$. Then $A_{i,\alpha}$ is t -closed in the remaining coordinates.*

Proof. If $y \notin A_{i,\alpha}$, then $(\alpha, y) \notin A$. Since A is closed, there exists $P \in I_t(A)$ with $P(\alpha, y) \neq 0$. The restriction $P|_{X_i=\alpha}$ vanishes on $A_{i,\alpha}$ and is nonzero at y . $\square$

Let A° denote the points of A lying on no full coordinate line contained in A . Then

$$A^\circ \cap \{X_i = \alpha\} \subseteq (A_{i,\alpha})^\circ. \quad (5)$$

Indeed, any full coordinate line in the slice is also a full coordinate line in A .

3. Interpolation from curve incidence

We now work over $k = \overline{\mathbb{F}_q}$. Here and below, $H^j(X, \mathcal{E})$ denotes sheaf cohomology, $\mathcal{I}_Z$ denotes the ideal sheaf of Z , and $\mathcal{E}(b) = \mathcal{E} \otimes \mathcal{O}_X(b)$. For a finite reduced set $Z \subseteq \mathbb{P}_k^a$ and an integral curve $C \subseteq \mathbb{P}_k^a$, let $\text{mult}_z C$ be the multiplicity of C at z , and define $\mu_Z(C) = \sum_{z \in Z \cap C} \text{mult}_z C$.

Lemma 3 (Seshadri interpolation). *Assume $\text{char } k = p > a$. Let $\Lambda \geq 0$, and suppose*

$$\mu_Z(C) \leq \Lambda \deg C \quad (6)$$

for every geometrically integral curve $C \subseteq \mathbb{P}^a$. Then

$$H^1(\mathbb{P}^a, \mathcal{I}_Z(b)) = 0 \quad (7)$$

whenever $b + a + 1 > a\Lambda$. In particular, arbitrary functions on Z interpolate in degree at most $\lceil a\Lambda \rceil$.

Proof. Let $\pi: Y = \text{Bl}_Z \mathbb{P}^a \rightarrow \mathbb{P}^a$, let E_z be the exceptional divisor above z , and write $H = \pi^* \mathcal{O}_{\mathbb{P}^a}(1)$ and $E = \sum_{z \in Z} E_z$. For the strict transform $C^\sharp$ of an integral curve C , $E \cdot C^\sharp = \mu_Z(C)$. Thus $\Lambda H - E$ is nef. Curves contained in an exceptional divisor also have nonnegative intersection with this divisor because $-E|_{E_z}$ is ample. By the blow-up description of multipoint Seshadri constants [8, Section 1] and Kleiman's ampleness criterion [7], it follows that $cH - E$ is ample for every $c > \Lambda$.

The canonical divisor is $K_Y = -(a+1)H + (a-1)E$. Therefore $bH - E = K_Y + ((b+a+1)H - aE)$. The divisor in parentheses is ample whenever $\frac{b+a+1}{a} > \Lambda$. The points of Z lift to disjoint sections over $W_2(k)$, so blowing up those sections gives a smooth $W_2(k)$ -lifting of Y . The field k is perfect, $\dim Y = a < p$, and the line bundle $(b+a+1)H - aE$ is ample. Deligne–Illusie Kodaira vanishing [3, Corollary 2.8] therefore yields $H^1(Y, \mathcal{O}_Y(bH - E)) = 0$. Moreover, $\pi_* \mathcal{O}_Y(-E) = \mathcal{I}_Z$ and $R^j \pi_* \mathcal{O}_Y(-E) = 0$ for $j > 0$. The Leray spectral sequence [9, Lemma 20.13.6] gives (7). $\square$

4. Hilbert growth of an expected component

Lemma 4 (Expected-component postulation). *For an integral projective variety $V \subseteq \mathbb{P}_k^a$, define its Hilbert function by*

$$H_V(b) = \dim_k \text{im} \left(H^0(\mathbb{P}^a, \mathcal{O}(b)) \rightarrow H^0(V, \mathcal{O}_V(b)) \right).$$

Let $X = V(E_1, \dots, E_c) \subseteq \mathbb{P}_k^a$ be the common zero locus of homogeneous forms of degrees at most $B \in \mathbb{Z}_{\geq 1}$. Let V be an irreducible component of X of the expected dimension $h = a - c \geq 1$, and put $\Delta = \deg V$. If $c = 0$, then $X = V = \mathbb{P}^a$, $\Delta = 1$, and the conclusion below holds with $R = 0$. Hence assume $c \geq 1$.

If $\text{char } k = p > c$, then there exists $R \leq cB$ such that

$$H_V(b) \geq \Delta \binom{b - R + h}{h} \quad (b \geq R), \quad (8)$$

If $h = 1$, let $\tilde{V} \rightarrow V$ be the normalization and let $g(\tilde{V})$ be its genus. Then

$$g(\tilde{V}) \leq cB\Delta + 1. \quad (9)$$

Proof. Choose a general complementary linear subspace $L \simeq \mathbb{P}^c$. Then L meets V transversely and $\Gamma = L \cap V$ is a reduced scheme consisting of Δ distinct smooth points. Moreover, L can be chosen so that Γ is disjoint from $V \cap W$ for every other irreducible component W of X .

Let $D \subseteq L$ be an integral curve meeting Γ . It cannot be contained in every $V(E_i)$. Otherwise $D \subseteq X$, so D lies in an irreducible component W of X . It cannot lie in V , since $L \cap V$ is finite, and it cannot lie in another component because $\Gamma \cap W = \emptyset$.

Thus some $E_i|_L$ cuts D properly. Weighted Bézout gives $\mu_\Gamma(D) \leq B \deg D$. Lemma 3 inside $L \simeq \mathbb{P}^c$ produces degree- R delta interpolants on Γ for some $R \leq cB$. Lift these forms to $\mathbb{P}^a$ using $H^0(\mathbb{P}^a, \mathcal{O}(R)) \twoheadrightarrow H^0(L, \mathcal{O}_L(R))$. Choose a hyperplane $\Pi_0 \subset L$ disjoint from Γ . Since $V \cap L = \Gamma$, $\Pi_0 \cap V = \emptyset$. Projection from Π_0 gives a finite morphism $\pi: V \rightarrow \mathbb{P}^h$. Indeed, every fiber lies in a $\mathbb{P}^c$ containing Π_0 as a hyperplane. A positive-dimensional projective fiber would meet Π_0 , which is impossible.

The degree of π is Δ , and one fiber is the reduced scheme Γ .

Let $k[V]$ be the homogeneous coordinate ring of V , and let $A = k[\ell_0, \dots, \ell_h] \subseteq k[V]$ be the resulting homogeneous Noether normalization. For a graded module M , write $M(-R)_b = M_{b-R}$ for its degree shift. Let $P_1, \dots, P_\Delta$ be the lifted degree- R delta interpolants. They define a graded A -linear map $A(-R)^\Delta \rightarrow k[V]$. Let $y_0 \in \mathbb{P}^h$ be the point whose reduced fiber is Γ , and dehomogenize near y_0 . After choosing a base linear form ℓ_0 nonzero there, the functions P_j/ℓ_0^R form a basis on Γ . Nakayama's lemma therefore makes the map surjective near y_0 . Both sides have generic rank Δ , so the map has full generic rank. Its kernel is a rank-zero submodule of a free module over the domain A , hence is zero. Thus $A(-R)^\Delta \hookrightarrow k[V]$. Taking degree- b pieces proves (8).

If $h = 1$, let $p_a(V)$ denote the arithmetic genus. For large b , $H_V(b) = \Delta b + 1 - p_a(V)$. Comparison with $H_V(b) \geq \Delta(b - R + 1)$ gives $p_a(V) \leq \Delta(R - 1) + 1 \leq cB\Delta + 1$. Normalization can only decrease genus, proving (9). □

5. Curve incidence for point strata

Define $\gamma_1 = 1$ and $\gamma_e = 1/(3 \cdot 6^{e-2})$ for $e \geq 2$.

Proposition 5. *There is a universal constant $c_0 \geq 1$ such that, for every $e \geq 1$ and $K \geq 1$, there are constants $P_{e,K} \geq 1$ and $0 < \theta_{e,K} \leq 1$ satisfying*

$$P_{e,K} \leq (c_0 e K)^{4e}, \quad \theta_{e,K}^{-\gamma_e} \leq (c_0 e K)^{4e}.$$

Write $[e] = \{1, \dots, e\}$ and set $m_0(e, K) = 2$. Let $S_1, \dots, S_e \leq \mathbb{F}_q^\times$ have pairwise coprime orders $n_i = |S_i|$, and assume that they are K -comparable, meaning $K^{-1} \leq n_i/n_j \leq K$ for all $i, j \in [e]$. Put $\Omega_e = \prod_{i=1}^e S_i$ and $m = \min_i n_i$. Let $A \subseteq \Omega_e$ be t -closed, where $m \geq m_0(e, K)$, $0 \leq t \leq \theta m$, and $0 < \theta \leq \theta_{e,K}$. Then every integral curve $C \subseteq \mathbb{P}_k^e$ satisfies

$$\boxed{\mu_{A^\circ}(C) \leq P_{e,K} \theta^{\gamma_e} m \deg C.} \tag{10}$$

Proof. We induct on e , choosing $P_{e,K}$ nondecreasing in e . If $t = 0$, a 0-closed set is either empty or the whole grid, and in both cases $A^\circ = \emptyset$. Hence we may assume $t \geq 1$, so $m\theta \geq 1$. Embed $\Omega_e \subseteq \mathbb{A}_k^e \subseteq \mathbb{P}_k^e$, where $\mathbb{A}_k^e$ is the chart $X_0 \neq 0$, and write $x_i = X_i/X_0$ for its affine coordinate functions.

For $e = 1$, either $A = S_1$, in which case $A^\circ = \emptyset$, or $A \neq S_1$. In the latter case, a nonzero locator in $I_t(A)$ has at least $|A|$ distinct zeros, so $|A^\circ| = |A| \leq t$. Now let $e \geq 2$, and assume the proposition in all smaller dimensions. Put $Z = A^\circ$. A curve contained in the hyperplane at infinity contains no points of Z .

If some affine coordinate x_i is constant on C , either its value is outside S_i , in which case $C \cap Z = \emptyset$, or Lemma 2 reduces the claim to $e - 1$ dimensions. After deleting this coordinate, let $m' = \min_{j \neq i} n_j$. Then $m \leq m' \leq Km$ and $t/m' \leq \theta$, so K -comparability absorbs the change.

Hence assume every affine coordinate is nonconstant on C .

If A is the whole grid, then $Z = \emptyset$. Otherwise choose a nonzero locator $F \in I_t(A)$. Its total degree is at most et . Let $E_1 = F^{\text{hom}}$ be its homogenization, so $\deg E_1 \leq et$.

If $C \not\subseteq V(E_1)$, weighted Bézout gives $\mu_Z(C) \leq et \deg C$, which is stronger than (10). Hence suppose $C \subseteq V(E_1)$.

Slicing estimate.

Put $S_{e,K} = K^e P_{e-1,K}$.

Let $V \subseteq \mathbb{P}^e$ be integral of dimension $h \geq 2$ and codimension $c = e - h$. Starting with V , process each current integral component W that meets Z separately. Choose an unused coordinate $i(W)$ that is nonconstant on W , and intersect W with all fibers $X_{i(W)} = \alpha$, $\alpha \in S_{i(W)}$. A previously sliced coordinate is constant, so the chosen coordinates are distinct along each branch. At one slicing level, Bézout gives

$$\sum_{\alpha \in S_{i(W)}} \deg U \leq n_{i(W)} \deg W \leq Km \deg W.$$

U an irreducible component of $W \cap \{X_{i(W)} = \alpha\}$; $U \cap Z \neq \emptyset$

After $h - 1$ slices, the terminal components are curves inside coordinate subspaces of dimension $e - (h - 1) = c + 1$. Their total degree is at most $(Km)^{h-1} \deg V$. All iterated slices of A are t -closed. Repeated use of (5) places every original point of Z in the point stratum of the corresponding terminal slice. The minimum subgroup order in a terminal grid lies between m and Km . Applying the induction hypothesis in dimension $c + 1$ gives

$$|Z \cap V| \leq S_{e,K} \theta^{\gamma_{c+1}} m^h \deg V. \quad (11)$$

Components at infinity contain no grid points.

Adapted separator flag.

Set $X_1 = V(E_1)$. Suppose inductively that $X_c = V(E_1, \dots, E_c)$ has the following properties, and put $B = \max_{1 \leq i \leq c} \deg E_i$:

1. Every irreducible component of X_c containing C is an actual component of dimension $e - c$. 2. Every E_i has degree at most B .

Let V be one such component, and put $h = e - c$, $j = c + 1$. Assume $h \geq 2$. Equation (11) gives

$$|Z \cap V| \leq S_{e,K} \theta^{\gamma_j} m^h \deg V. \quad (12)$$

Lemma 4 gives

$$H_V(b) \geq \deg V \binom{b - cB + h}{h}. \quad (13)$$

Since $\binom{x+h}{h} \geq x^h/h!$ and $m\theta^{\gamma_j/h} \geq m\theta \geq 1$, after rounding upward there is some b in the range

$$cB \leq b \leq cB + 2(h!S_{e,K})^{1/h} m\theta^{\gamma_j/h}, \quad (14)$$

such that $H_V(b) > |Z \cap V|$. Linear algebra produces a homogeneous form P_V such that

$$P_V|_{Z \cap V} = 0, \quad P_V|_V \neq 0. \quad (15)$$

If $P_V|_C \neq 0$, then P_V vanishes at every point of $Z \cap C$. Let $i_z(C, P_V)$ denote the local intersection multiplicity of C and $V(P_V)$ at z . At each such point, $i_z(C, P_V) \geq \text{mult}_z C$, so Bézout gives $\mu_Z(C) \leq (\deg P_V) \deg C$. Otherwise construct P_V for every irreducible component V of X_c containing C , and suppose every P_V contains C . Multiply by powers of linear forms not vanishing identically on the tracked components so that the resulting forms all have the same degree.

A generic linear combination E_{c+1} :

- vanishes identically on C ; - does not vanish identically on any tracked component.

Indeed, for each tracked component the condition that the linear combination vanish identically is a proper linear subspace of the coefficient space.

Thus E_{c+1} cuts every tracked component properly. Every component of $X_{c+1} := X_c \cap V(E_{c+1})$ containing C lies in a tracked component of X_c . Krull's principal ideal theorem then shows that it has dimension exactly $e - c - 1$. This propagates the induction invariant.

Since $h \geq 2$, the second term in (14) is at most $2(e!S_{e,K})^{1/2}m\theta^{\gamma_j/h}$. The degree recurrence therefore gives the following bound with

$$H_{e,K} = 4e^e(e!S_{e,K})^{1/2}.$$

For $e = 2$, set $\beta_2 = 1$. For $e \geq 3$, set

$$\beta_e = \min_{2 \leq j \leq e-1} \frac{\gamma_j}{e - j + 1}. \quad (16)$$

For the chosen sequence, $2 \cdot 6^{e-1-j} \geq e - j + 1$, and hence

$$\beta_e = \frac{\gamma_{e-1}}{2} \quad (e \geq 3). \quad (17)$$

Either an earlier Bézout alternative completes the proof, or C becomes an expected component of $e - 1$ hypersurfaces whose degrees are bounded by

$$B_* \leq H_{e,K}m\theta^{\beta_e}. \quad (18)$$

For $e = 2$, no intermediate separator is needed: C is already a component of the initial locator, and $B_* \leq 2t \leq H_{2,K}m\theta$.

Every earlier Bézout branch gives $\mu_Z(C) \leq H_{e,K}m\theta^{\beta_e} \deg C$, which is stronger than the final required bound because $0 < \theta < 1$.

Arithmetic on the final curve.

Let $\eta: \tilde{C} \rightarrow C$ be the normalization, and let $k(C)$ be the function field of C . For a nonconstant $f \in k(C)$, write $\deg f$ for the degree of the induced map $\tilde{C} \rightarrow \mathbb{P}^1$, equivalently the total pole order of f , and write df for its differential. Let $D = \deg C$. Bézout and Lemma 4 give

$$D \leq B_*^{e-1}, \quad g(\tilde{C}) \leq (e-1)B_*D + 1. \quad (19)$$

Choose θ so that $H_{e,K}\theta^{\beta_e} \leq 1/2$. Then $B_* < m$. By coprimality, $N \mid q - 1$, so $D < m^e \leq N < q$.

Each coordinate function x_i has degree at most D . Since x_i is nonconstant and $\deg x_i < q$, its differential is nonzero: in characteristic q , a rational function with zero differential is a q -th power, whose nonzero degree is divisible by q .

Put $u_i = x_i^{n_i}$. Since $q \nmid n_i$, every $du_i \neq 0$.

Multiplicatively independent case.

Two functions are multiplicatively independent modulo constants if their classes in $k(C)^\times/k^\times$ are independent: $u_i^a u_j^b \in k^\times$ for integers a, b implies $a = b = 0$. Suppose some pair u_i, u_j has this property.

For a point $P \in \tilde{C}$ and $f \in k(C)^\times$, let $\text{ord}_P(f)$ be its order of vanishing at P , with negative values at poles. Let Σ be the union of the zero and pole supports of u_i and u_j , and define

$$\mathcal{G}(u_i, u_j) = \sum_{P \in \tilde{C} \setminus \Sigma} \min\{\text{ord}_P(1 - u_i), \text{ord}_P(1 - u_j)\}. \quad (20)$$

Equivalently,

$$\mathcal{G}(u_i, u_j) = \sum_{P \in \tilde{C}} \min\{\text{ord}_P^+(1 - u_i), \text{ord}_P^+(1 - u_j)\}, \quad (21)$$

where $\text{ord}_P^+(f) = \max\{0, \text{ord}_P(f)\}$. Thus poles contribute zero.

Set $\chi = \max\{1, |\Sigma| + 2g(\tilde{C}) - 2\}$. The positive-characteristic Corvaja–Zannier estimate [2, Theorem 2] gives

$$\mathcal{G}(u_i, u_j) \leq \max\left\{3\sqrt[3]{2}(\deg u_i \deg u_j \chi)^{1/3}, 12 \frac{\deg u_i \deg u_j}{q}\right\}, \quad (22)$$

Let $\xi \in C \cap \Omega$, and let $\tilde{\xi} \in \eta^{-1}(\xi)$ be a normalization branch above ξ . Since $\xi_i^{n_i} = 1$,

$$x_i^{n_i} - 1 = (x_i - \xi_i) \left(x_i^{n_i-1} + x_i^{n_i-2} \xi_i + \cdots + \xi_i^{n_i-1} \right).$$

The second factor evaluates at ξ to $n_i \xi_i^{n_i-1} \neq 0$. Hence

$$\text{ord}_{\tilde{\xi}}(1 - u_i) = \text{ord}_{\tilde{\xi}}(x_i - \xi_i), \quad (23)$$

and similarly for j . The functions $x_a - \xi_a$ generate the maximal ideal of C at ξ . For a reduced curve, multiplicity is the sum over normalization branches of the valuation of this maximal ideal. Therefore

$$\text{mult}_{\xi} C = \sum_{\tilde{\xi} \in \eta^{-1}(\xi)} \min_{1 \leq a \leq e} \text{ord}_{\tilde{\xi}}(x_a - \xi_a). \quad (24)$$

Every grid branch lies outside Σ , so

$$\text{mult}_{\xi} C \leq \sum_{\tilde{\xi} \in \eta^{-1}(\xi)} \min\{\text{ord}_{\tilde{\xi}}(1 - u_i), \text{ord}_{\tilde{\xi}}(1 - u_j)\}.$$

Summing over $\xi \in Z \cap C$,

$$\boxed{\mu_Z(C) \leq \mathcal{G}(u_i, u_j)}. \quad (25)$$

We have $\deg u_i, \deg u_j \leq KmD$ and $|\Sigma| \leq 4D$. By (19), $\chi \leq 8eB_*D$. For the second term in (22), use $D \leq B_*^{e-1}$ and $q > m^e$. Therefore

$$\mu_Z(C) \leq 12eK^2m \left[\left(\frac{B_*}{m} \right)^{1/3} + \left(\frac{B_*}{m} \right)^{e-1} \right] D. \quad (26)$$

Multiplicatively dependent case.

Suppose every pair u_i, u_j is multiplicatively dependent modulo constants.

The group $k(C)^\times / k^\times$ is torsion-free. Hence the subgroup generated by the classes of the x_i has rank one. There exist $z \in k(C)^\times$, constants $\lambda_i \in k^\times$, and nonzero integers a_i such that

$$x_i = \lambda_i z^{a_i}, \quad \gcd_i a_i = 1. \quad (27)$$

Since the affine coordinates generate $k(C)$, $k(C) = k(z)$. Also, $|a_i| \leq \deg x_i \leq D < q$. A grid point must satisfy $z^{n_i a_i} = \lambda_i^{-n_i}$, for every i . The number of simultaneous solutions is at most

$$g_0 = \gcd_i(n_i |a_i|). \quad (28)$$

For a prime ℓ and a positive integer r , let $v_\ell(r)$ be the exponent of ℓ in r . At most one n_i is divisible by ℓ . Thus at least $e - 1$ of the a_i have ℓ -adic valuation at least $v_\ell(g_0)$. Consequently, $g_0^{e-1} \mid \prod_i |a_i|$. Therefore

$$g_0 \leq \left(\prod_i |a_i| \right)^{1/(e-1)} \leq D^{e/(e-1)} \leq B_* D. \quad (29)$$

Choose integers b_i with $\sum_i b_i a_i = 1$. Then $z = \lambda \prod_i x_i^{b_i}$ for a constant λ . Write $\mathbb{G}_m = \mathbb{A}_k^1 \setminus \{0\}$ for the multiplicative group. The primitive monomial map

$$\mathbb{G}_m \longrightarrow \mathbb{G}_m^e, \quad z \longmapsto (\lambda_1 z^{a_1}, \dots, \lambda_e z^{a_e})$$

is a closed immersion whose image is exactly $C \cap \mathbb{G}_m^e$. This image is smooth, so every grid point has multiplicity one. Hence

$$\mu_Z(C) \leq g_0 \leq B_* D. \quad (30)$$

Since $B_*/m \leq 1/2$, equations (26) and (30), together with the earlier Bézout alternative, give (10) with

$$P_{1,K} = 1, \quad P_{e,K} = \max\{K P_{e-1,K}, 32eK^2 H_{e,K}\}.$$

Define $\theta_{1,K} = 1$ and, for $e \geq 2$,

$$\theta_{e,K} = \min \left\{ \frac{1}{2}, \theta_{e-1,K}, (2H_{e,K})^{-1/\beta_e} \right\}.$$

For $e = 2$, $\gamma_2 = \beta_2/3 = 1/3$. For $e \geq 3$, (17) gives $\gamma_e = \beta_e/3 = \gamma_{e-1}/6$. Thus

$$\boxed{\gamma_e = \frac{1}{3 \cdot 6^{e-2}}}.$$

Finally, induction in the two displayed recurrences, using $e! \leq e^e$, gives $P_{e,K} \leq (c_0 e K)^{4e}$ for a universal c_0 . The same induction gives $\theta_{e,K}^{-\gamma_e} \leq (c_0 e K)^{4e}$: the new smallness condition costs only $(2H_{e,K})^{\gamma_e/\beta_e} = (2H_{e,K})^{1/3}$, while the inherited condition is raised to the power $\gamma_e/\gamma_{e-1} \leq 1/3$. Since $m \geq 2$ implies $q > N \geq 2^e > e$, it also supplies every characteristic hypothesis used above. This proves Proposition 5 with $m_0(e, K) = 2$. □

6. Interpolation of the point stratum

Under the hypotheses of Proposition 5, Lemma 3 shows that arbitrary functions on A° interpolate in ordinary degree

$$R_e \leq \lceil e P_{e,K} \theta^{\gamma_e} m \rceil \leq e P_{e,K} \theta^{\gamma_e} m + 1. \quad (31)$$

An affine polynomial of total degree R_e has individual degree at most R_e . Separate homogenization therefore gives interpolation on A° by sections of $\mathcal{O}(R_e, \dots, R_e)$ on $(\mathbb{P}^1)^e$.

7. Maximal coordinate flats

Let $T \subsetneq \Omega$ be t -closed. For $J \subseteq [d]$, write $S_J = \prod_{i \in J} S_i$. For $b \in \prod_{i \notin J} S_i$, call $S_J \times \{b\}$ a full J -flat. For $J \subsetneq [d]$, define

$$F_J = \left\{ b \in \prod_{i \notin J} S_i : S_J \times \{b\} \subseteq T \right\}. \quad (32)$$

Then F_J is t -closed.

Indeed, if $b \notin F_J$, choose $x_J \in S_J$ such that $(x_J, b) \notin T$. A polynomial in $I_t(T)$ separating (x_J, b) from T , after fixing the variables indexed by J to x_J , vanishes on F_J and is nonzero at b .

Put

$$B_J = F_J^\circ. \quad (33)$$

Then $b \in B_J$ exactly when $S_J \times \{b\}$ is a maximal full coordinate flat in T . Indeed, a full coordinate line through b inside F_J is equivalent to enlarging the corresponding J -flat by that coordinate.

Put $L_{d,K} := KdP_{d,K} \leq Kd(c_0dK)^{4d}$. Applying (31) in dimension $d - |J|$, and using $\gamma_{d-|J|} \geq \gamma_d$, shows that every nonempty B_J interpolates in a common individual degree

$$R \leq L_{d,K} \theta^{\gamma_d} m + 1. \quad (34)$$

Take $R = \lceil L_{d,K} \theta^{\gamma_d} m \rceil$, increasing it to 1 if necessary. The bound (34) remains valid.

8. Coordinate-flat arrangement regularity

Let $X_e = (\mathbb{P}^1)^e \hookrightarrow \mathbb{P}^{s_e}$, where $s_e = 2^e - 1$, be the Segre embedding. For $J \subseteq [e]$, write $J^c = [e] \setminus J$, let $B_J \subseteq (\mathbb{P}^1)^{J^c}$ be a finite reduced set, and put $Y_J = (\mathbb{P}^1)^J \times B_J$. We call J its orientation and each $(\mathbb{P}^1)^J \times \{b\}$ a J -flat. Let $Y = \bigcup_{J \subseteq [e]} Y_J$ be the reduced union, where every B_J interpolates in individual degree R . Increasing R if necessary, assume $R \geq 1$.

Let $S = k[z_U \mid U \subseteq [e]]$ be the homogeneous coordinate ring of the ambient $\mathbb{P}^{s_e}$, with variables indexed by the Segre coordinates. For a closed subscheme $Z \subseteq \mathbb{P}^{s_e}$, let $I_Z \subseteq S$ be its saturated homogeneous ideal, and let S_n be the degree- n part of S . Write reg for Castelnuovo–Mumford regularity of a graded S -module or its associated sheaf.

Before doing anything else, discard every orientation J for which $B_J = \emptyset$. This does not change Y . Let $\mathcal{F} = \{J \mid B_J \neq \emptyset\}$. If $\mathcal{F} = \emptyset$, then $Y = \emptyset$, and all regularity and extension statements below are immediate. Henceforth every Y_J is nonempty, so $I_{Y_J} \subsetneq S$ is a proper homogeneous ideal.

Lemma 6 (Ambient regularity). *For $e \geq 1$, put*

$$c_e := 1 + 4^e(2^{e-1}(2^e - 1) - 1) < 2^{4e-1}.$$

Then

$$\boxed{\text{reg}_{\mathbb{P}^{s_e}} I_Y \leq c_e(R + 1)}. \quad (35)$$

Proof. **One orientation.**

Suppose first that $\mathcal{F} = \{J\}$.

Interpolation on B_J propagates from degree R to every $n \geq R$. Indeed, choose a section $u \in H^0((\mathbb{P}^1)^{J^c}, \mathcal{O}(1, \dots, 1))$ nonzero at every point of B_J . Given target values in degree n , divide them by u^{n-R} , interpolate in degree R , and multiply back by u^{n-R} .

It follows that

$$H^0(X_e, \mathcal{O}(n, \dots, n)) \twoheadrightarrow H^0(Y_J, \mathcal{O}_{Y_J}(n)).$$

Segre projective normality lifts these sections to ambient degree- n forms. Moreover, Künneth gives $H^j(Y_J, \mathcal{O}_{Y_J}(n)) = 0$ for $j > 0$ and $n \geq -1$. The ambient ideal sequence therefore gives

$$\text{reg } I_{Y_J} \leq \max\{R + 1, e\}. \quad (36)$$

In particular, I_{Y_J} is generated in degrees at most $\max\{R + 1, e\}$.

We will also use the following product consequence. If $Z \subseteq (\mathbb{P}^1)^r$ has ambient Segre ideal of regularity a , then, for fixed total dimension $e = r + \ell$,

$$\text{reg } I_{(\mathbb{P}^1)^\ell \times Z} \leq \max\{a, e\}. \quad (37)$$

To see this, ambient regularity of I_Z gives:

- surjectivity onto $H^0(Z, \mathcal{O}_Z(n))$ for $n \geq a - 1$; - vanishing of $H^j(Z, \mathcal{O}_Z(n))$ for $j > 0$ in the corresponding regularity range.

Segre projective normality gives the required global-section surjectivity for the product, and Künneth gives the higher-cohomology vanishings. Checking the twists in the definition of regularity gives (37).

Induction on dimension and orientations.

Induct lexicographically on e and $f = |\mathcal{F}|$. The cases $f = 0$ and $f = 1$ have been handled.

If $[e] \in \mathcal{F}$, then $B_{[e]}$ is the unique point of the zero-dimensional complementary product, so $Y_{[e]} = X_e$. Hence $Y = X_e$, and the conclusion is immediate.

Assume now that $[e] \notin \mathcal{F}$. Let $L = \bigcap_{J \in \mathcal{F}} J$. If $L \neq \emptyset$, put $e' = e - |L|$. Then $Y = (\mathbb{P}^1)^L \times \bar{Y}$, where $\bar{Y}$ is an e' -dimensional coordinate-flat arrangement with the same interpolation parameter R . Lower-dimensional induction and (37) apply.

Thus assume

$$\bigcap_{J \in \mathcal{F}} J = \emptyset. \quad (38)$$

Write $\mathfrak{m} = (z_U \mid U \subseteq [e])$, $I = I_Y$, and $M = S/I$.

For each $J \in \mathcal{F}$, set

$$Y^{(J)} = \bigcup_{\substack{J' \in \mathcal{F} \\ J' \neq J}} Y_{J'}, \quad M_J = S/I_{Y^{(J)}}.$$

There is a quotient map $M \twoheadrightarrow M_J$. Its kernel is annihilated by the proper ideal I_{Y_J} , because

$$I_{Y_J} I_{Y^{(J)}} \subseteq I_{Y_J} \cap I_{Y^{(J)}} = I_Y. \quad (39)$$

Condition (38) implies that $\mathcal{C} = \bigcap_{J \in \mathcal{F}} Y_J$ is finite: every coordinate is fixed by at least one orientation.

For $c \in \mathcal{C}$, let Y_c be the union of the individual component flats passing through c .

For each $J \in \mathcal{F}$, degree- R interpolation on B_J supplies a selector for the unique J -flat through c , vanishing on every other J -flat. Tensor this selector with a degree- R section in each free coordinate that is nonzero at c , and use Segre projective normality to obtain an ambient homogeneous form

$e_{J,c} \in S_R$ such that $e_{J,c}(c) \neq 0$ and $e_{J,c}$ vanishes identically on every other component of orientation J .

Set $h_c = \prod_{J \in \mathcal{F}} e_{J,c}$. Then

$$h_c(c) \neq 0, \quad \deg h_c = fR \leq 2^e R. \quad (40)$$

Every component flat not containing c is annihilated by one factor of h_c . Therefore

$$h_c I_{Y_c} \subseteq I_Y. \quad (41)$$

Consequently, the kernel of the quotient map $M \twoheadrightarrow M_c := S/I_{Y_c}$ is annihilated by the proper homogeneous ideal (h_c) .

Uniform regularity of the stars.

Move c to $([1 : 0], \dots, [1 : 0])$, and put $\mathcal{D} = \bigcup_{J \in \mathcal{F}} 2^J$, where 2^J denotes the set of subsets of J . For the usual Hibi sorting order [5], let $\text{in}(I)$ denote the initial ideal of a homogeneous ideal $I \subseteq S$. Then

$$\text{in}(I_{Y_c}) = (z_U \mid U \notin \mathcal{D}) + (z_U z_V : U, V \in \mathcal{D} \text{ incomparable}). \quad (42)$$

Indeed, if $U \cup V \in \mathcal{D}$, then the Segre relation $z_U z_V - z_{U \cap V} z_{U \cup V}$ lies in I_{Y_c} and has leading term $z_U z_V$. If $U \cup V \notin \mathcal{D}$, no component of Y_c supports both coordinates, so $z_U z_V \in I_{Y_c}$.

The standard monomials are precisely the chain monomials

$$z_{U_1} \cdots z_{U_n}, \quad U_1 \subseteq \cdots \subseteq U_n, \quad U_n \in \mathcal{D}.$$

They are linearly independent modulo I_{Y_c} . Choose J with $U_n \subseteq J$ and restrict to the corresponding J -flat. The resulting coordinate monomial records the number of sets U_r containing each i , and these counts uniquely recover the chain.

Thus (42) is the full initial ideal. It is the Stanley–Reisner ideal of the order complex of $\mathcal{D}$. Every chain has at most $e + 1$ elements. The Gröbner-degeneration inequality [1] and Hochster’s formula [6] therefore give

$$\text{reg } M_c \leq \text{reg}(S/\text{in}(I_{Y_c})) \leq e + 1. \quad (43)$$

The approximation system.

Define

$$\mathfrak{b} = \sum_{J \in \mathcal{F}} I_{Y_J} + \sum_{c \in \mathcal{C}} (h_c). \quad (44)$$

If $\mathcal{C} = \emptyset$, omit the second sum.

The projective zero locus of $\mathfrak{b}$ is empty. Indeed, a common zero of every I_{Y_J} must lie in $\mathcal{C}$, whereas for each $c \in \mathcal{C}$, $h_c(c) \neq 0$. In the present branch $2 \leq f \leq 2^e - 1$. Every I_{Y_J} is generated in degrees at most $\max\{R + 1, e\}$, and $\deg h_c = fR$. Thus all generators have degree at most

$$D_0 = \max\{e, fR\}.$$

Let $\mathfrak{b}_{D_0} = \mathfrak{b} \cap S_{D_0}$. This degree- D_0 subsystem is basepoint-free: if a lower-degree generator is nonzero at a point, multiply it by a complementary-degree form nonzero there.

Since k is infinite, $s_e + 1$ general elements of $\mathfrak{b}_{D_0}$ have no common projective zero. They form a regular sequence in S , hence an Artinian complete intersection with socle degree $(s_e + 1)(D_0 - 1)$. Therefore

$$\mathfrak{m}^\tau \subseteq \mathfrak{b}, \quad \tau = (s_e + 1)(D_0 - 1) + 1. \quad (45)$$

The modules $\{M_J\}_{J \in \mathcal{F}} \cup \{M_c\}_{c \in \mathcal{C}}$ form a degree- τ approximation system for M :

- $M \twoheadrightarrow M_J$ has kernel annihilated by the proper ideal I_{Y_J} ; - $M \twoheadrightarrow M_c$ has kernel annihilated by the proper ideal (h_c) ; - the sum of these annihilator ideals contains $\mathfrak{m}^\tau$.

Let

$$b_0 = \max(\{0\} \cup \{\text{reg } M_J : J \in \mathcal{F}\} \cup \{\text{reg } M_c : c \in \mathcal{C}\}).$$

Apply Derksen–Sidman’s approximation theorem [4, Theorem 3.5] with $r = b_0 + 1 + (\tau - 1)(s_e + 1)$. The module M is generated in degree 0, and $r - (\tau - 1)(s_e + 1) = b_0 + 1$, while every approximating module has regularity at most $b_0 = r - (\tau - 1)(s_e + 1) - 1$. Thus

$$\text{reg } M \leq b_0 + 1 + (\tau - 1)(s_e + 1). \quad (46)$$

Define

$$\mathcal{R}_{e,f}(R) = \max\{R, e + 1\} + \sum_{j=2}^f [1 + 4^e(\max\{e, jR\} - 1)].$$

Lexicographic induction on (e, f) gives $\text{reg } M \leq \mathcal{R}_{e,f}(R)$. In the product branch, lower-dimensional induction and (37) give $\text{reg } M \leq \max\{\mathcal{R}_{e',f}(R), e - 1\} \leq \mathcal{R}_{e,f}(R)$. In the branch (38), every M_J has regularity at most $\mathcal{R}_{e,f-1}(R)$, every M_c has regularity at most $e + 1$, and (46) gives the claimed bound.

Since $[e]$ was handled separately, $f \leq 2^e - 1$. Put $\sigma_e = \sum_{j=2}^{2^e-1} j = 2^{e-1}(2^e - 1) - 1$. For $e \geq 2$,

$$\mathcal{R}_{e,2^e-1}(R) + 1 \leq (1 + 4^e \sigma_e)R + e + 2^e + 4^e(2^e - 2)(e - 1) \leq c_e(R + 1).$$

The last inequality follows from $4^e[\sigma_e - (2^e - 2)(e - 1)] \geq e + 2^e - 1$. The case $e = 1$ follows directly from (36). This proves Lemma 6. $\square$

9. Unequal multidegrees

Since $c_e \leq c_d$ for $e \leq d$, put $a_d(R) = c_d(R + 1)$ and $b = a_d(R) - 1$. For a sheaf $\mathcal{E}$ on X_e and $\mathbf{v} = (v_1, \dots, v_e)$, write $\mathcal{E}(\mathbf{v}) = \mathcal{E} \otimes_{\mathcal{O}_{X_e}} \mathcal{O}_{X_e}(v_1, \dots, v_e)$, and let $\mathcal{I}_{Y/X_e}$ denote the ideal sheaf of Y in X_e .

Lemma 7 (Anisotropic extension). *For every $e \leq d$, every coordinate-flat arrangement $Y \subseteq X_e$, and every multidegree $\mathbf{v} = (v_1, \dots, v_e)$, $v_i \geq b$, one has*

$$H^1(X_e, \mathcal{I}_{Y/X_e}(\mathbf{v})) = 0. \quad (47)$$

Proof. For $e = 0$, $X_0 = \text{Spec } k$, so every H^1 vanishes. This is the dimension-zero base case.

The cases $Y = \emptyset$ and $Y = X_e$ are immediate, so assume Y is nonempty and proper.

Ambient $a_d(R)$ -regularity gives $H^1(\mathbb{P}^{s_e}, \mathcal{I}_{Y/\mathbb{P}^{s_e}}(b)) = 0$. Thus

$$H^0(\mathbb{P}^{s_e}, \mathcal{O}(b)) \longrightarrow H^0(Y, \mathcal{O}_Y(b))$$

is surjective. This map factors through $H^0(X_e, \mathcal{O}_{X_e}(b, \dots, b))$. Moreover, Künneth gives $H^1(X_e, \mathcal{O}_{X_e}(b, \dots, b)) = 0$ because $b \geq 0$. Hence the intrinsic ideal sequence gives

$$H^1(X_e, \mathcal{I}_{Y/X_e}(b, \dots, b)) = 0. \quad (48)$$

We induct first on e , and for fixed e on the excess $\text{ex}(\mathbf{v}) = \sum_i (v_i - b)$. The case $\text{ex}(\mathbf{v}) = 0$ is (48).

Suppose $\text{ex}(\mathbf{v}) > 0$, and choose i with $v_i > b$. Let $\Sigma_i \subseteq \mathbb{P}^1(k)$ be the finite set of all i -coordinates fixed on the component flats of Y . Choose $\beta \notin \Sigma_i$, and let $H \subseteq X_e$ be the fiber of the i -th projection over β , so $H \simeq X_{e-1}$. No irreducible component of Y is contained in H . Since Y is reduced, the defining equation of H lies in no minimal prime of $\mathcal{O}_Y$, hence is a nonzerodivisor. Equivalently, $\text{Tor}_1^{\mathcal{O}_{X_e}}(\mathcal{O}_Y, \mathcal{O}_H) = 0$. Let $\mathbf{e}_i$ be the i -th standard basis vector, and let $\mathbf{v}_{\hat{i}}$ be obtained from $\mathbf{v}$ by deleting its i -th coordinate. Tensoring the intrinsic ideal sequence with $\mathcal{O}_H$, and using the Cartier-divisor sequence for H , gives an exact sequence

$$0 \longrightarrow \mathcal{I}_{Y/X_e}(\mathbf{v} - \mathbf{e}_i) \longrightarrow \mathcal{I}_{Y/X_e}(\mathbf{v}) \longrightarrow \mathcal{I}_{Y \cap H/H}(\mathbf{v}_{\hat{i}}) \longrightarrow 0. \quad (49)$$

The slice is again a coordinate-flat arrangement scheme-theoretically. Indeed, on $U = \mathbb{P}^1 \setminus \Sigma_i$, every component with fixed i -coordinate disappears, while every component free in coordinate i is the product of U with its projection. Thus

$$Y|_{U \times X_{e-1}} = U \times \bigcup_{J \ni i} (\mathbb{P}^1)^{J \setminus \{i\}} \times B_J.$$

In particular,

$$Y \cap H = \bigcup_{J \ni i} (\mathbb{P}^1)^{J \setminus \{i\}} \times B_J$$

as schemes. It is reduced and has the same interpolation parameter R .

The first term of (49) has vanishing H^1 by induction on $\text{ex}(\mathbf{v})$, and the last term has vanishing H^1 by induction on e . The long exact sequence proves (47). $\square$

10. Local-to-global interpolation on T

Return to $X = (\mathbb{P}^1)^d$. For each $J \subsetneq [d]$ with $B_J \neq \emptyset$, define

$$Y_J = (\mathbb{P}^1)^J \times B_J, \quad Y = \bigcup_{B_J \neq \emptyset} Y_J. \quad (50)$$

The grid points of Y are exactly those of T , namely $Y(k) \cap \Omega = T$.

Let $f: T \rightarrow k$ satisfy

$$f|_L \in R_i \otimes_{\mathbb{F}_q} k \quad (51)$$

on every full direction- i line $L \subseteq T$.

Consider a maximal J -flat $S_J \times \{b\} \subseteq T$. Every coordinate-line constraint in the free directions J holds. We claim that the restriction of f to this flat is represented by a tensor-product polynomial with

$$\deg_{X_i} < r_i \quad (i \in J). \quad (52)$$

Proceed by induction on $|J|$. Choose $i \in J$, and write $y = (x_j)_{j \in J \setminus \{i\}}$ for the remaining free coordinates. A fixed left inverse to the Vandermonde evaluation map expresses $f(x_i, y) = \sum_{a < r_i} x_i^a c_a(y)$. Each coefficient $c_a(y)$ is a fixed linear combination of the values $f(x_i, y)$ as x_i varies. Therefore it retains every line constraint in the remaining directions. Apply induction.

Thus each maximal flat carries a section of $\mathcal{O}_X(r_1 - 1, \dots, r_d - 1)$. These sections agree on pairwise intersections. If maximal flats with free-coordinate sets J_1 and J_2 meet, their intersection is a J' -flat for $J' = J_1 \cap J_2$. Their sections agree on its full subgroup grid because both equal f . Since $r_i \leq n_i$ for $i \in J'$, tensor Vandermonde uniqueness implies equality on the whole projective J' -flat.

These compatible sections glue to a section on Y . Indeed, after trivializing the line bundle near a point of Y , the incident component ideals are coordinate monomial primes. Such ideals form a distributive lattice under sums and intersections, so a tuple of sections agreeing modulo every pairwise sum comes from the quotient by their intersection.

Assume $\theta \leq \theta_{d,K}$ and

$$c_d L_{d,K} \theta^{\gamma_d} \leq \frac{\varepsilon}{2}. \quad (53)$$

Equations (34) and (53) give $a_d(R) - 1 \leq \varepsilon m - 1$ whenever $m \geq 4c_d/\varepsilon$.

But $r_i - 1 \geq \varepsilon n_i - 1 \geq \varepsilon m - 1$. Lemma 7 therefore extends the section on Y to a global section of $\mathcal{O}_X(r_1 - 1, \dots, r_d - 1)$. Dehomogenizing gives $P \in \mathcal{B}_{\mathbf{r}} \otimes_{\mathbb{F}_q} k$ such that $P|_T = f$. Hence

$$\boxed{\text{ev}_T(\mathcal{B}_{\mathbf{r}} \otimes k) = \{f : T \rightarrow k : f|_L \in R_i \otimes k \text{ on every full direction-}i \text{ line}\}.} \quad (54)$$

Both sides are scalar extensions of subspaces defined over $\mathbb{F}_q$. Equality therefore descends to $\mathbb{F}_q$.

11. Duality and supported-line decomposition

For every full direction- i line $L \subseteq T$, write $\text{dir}(L) = i$, and let $C_i[L] \subseteq \mathbb{F}_q^T$ be the subspace of vectors supported on L whose restriction to L lies in C_i .

Inside $\mathbb{F}_q^T$, the orthogonal complement of

$$\sum_{\substack{L \subseteq T \\ L \text{ full}}} C_{\text{dir}(L)}[L]$$

is the space of functions whose restriction to every full direction- i line lies in $C_i^\perp = R_i$. By (54), this orthogonal complement equals

$$\text{ev}_T(\mathcal{B}_{\mathbf{r}}). \quad (55)$$

Let $M \in Q_1 + \dots + Q_d$ be supported in T . By (2), M is orthogonal to $\text{ev}_\Omega(\mathcal{B}_{\mathbf{r}})$. Since M vanishes outside T , it is orthogonal to every restriction in (55). Double orthogonality gives

$$M = \sum_{\substack{L \subseteq T \\ L \text{ full}}} M_L, \quad M_L \in C_{\text{dir}(L)}[L]. \quad (56)$$

Group by direction: $A_i = \sum_{\text{dir}(L)=i} M_L$. Then $A_i \in Q_i$ and $M = A_1 + \dots + A_d$. For fixed i , direction- i lines are disjoint and every active one is contained in T . Therefore $n_i |A_i|_i \leq |T|$. Summing over i ,

$$\boxed{\sum_{i=1}^d n_i |A_i|_i \leq d|T|.} \quad (57)$$

12. Product expansion

Put

$$\mathcal{A}_{d,K} = \max\{2c_d L_{d,K}, \theta_{d,K}^{-\gamma_d}, 2\}, \quad B_d = 8c_d, \quad \theta = \left(\frac{\varepsilon}{\mathcal{A}_{d,K}}\right)^{1/\gamma_d}.$$

Then $\theta \leq \min\{\varepsilon, \theta_{d,K}\}$, and (53) holds. Moreover,

$$\mathcal{A}_{d,K} B_d^{\gamma_d} \leq (C_0 d K)^{10d} \quad (58)$$

for a universal C_0 , by $c_d < 2^{4d-1}$, Proposition 5, and (34). Indeed, for another universal constant C , one has $\mathcal{A}_{d,K} \leq (CdK)^{5d}$ and $B_d^{\gamma_d} \leq (CdK)^d$, which is stronger than (58).

Suppose first that $m < B_d/\theta$. For $M = 0$, use the zero decomposition. If $M \neq 0$, choose any decomposition $M = \sum_i A_i$ with $A_i \in Q_i$. Since there are at most N/n_i direction- i lines and $N \leq K^d m^d$,

$$\sum_i n_i |A_i|_i \leq dN < dK^d \left(\frac{B_d}{\theta}\right)^d |M|.$$

Thus this range has product-expansion constant $\theta^d/(dK^d B_d^d)$. Hence assume $m \geq B_d/\theta$. This implies $m \geq 4c_d/\varepsilon$, $t = \lfloor \theta m \rfloor \geq 1$, and all hypotheses of the local-to-global argument.

If $M = 0$, take every $A_i = 0$. Assume $M \neq 0$. Choose $t = \lfloor \theta m \rfloor$ and $u = t + 1$, and set $W = \text{supp } M$, $T = \text{cl}_t(W)$. By Lemma 1,

$$|T| \leq \frac{N}{u^d} |M| \leq \left(\frac{K}{\theta}\right)^d |M|. \quad (59)$$

Set

$$\delta = \frac{\theta^d}{2K^d}. \quad (60)$$

First suppose $|M| < \delta N$. Since $N \leq K^d m^d$, $|M| < \frac{\theta^d m^d}{2} < u^d$. The space $\mathcal{P}_t$ has dimension u^d , so there is a nonzero locator vanishing on W . Because $t < n_i$, evaluation of $\mathcal{P}_t$ on Ω is injective. Thus this locator cannot vanish on all of Ω , and $T \subsetneq \Omega$. The local-to-global theorem applies. Combining (57) and (59),

$$\sum_i n_i |A_i|_i \leq d \left(\frac{K}{\theta}\right)^d |M|. \quad (61)$$

Now suppose $|M| \geq \delta N$. Choose any decomposition $M = A_1 + \dots + A_d$, $A_i \in Q_i$. There are at most N/n_i direction- i lines, so

$$\sum_i n_i |A_i|_i \leq dN \leq \frac{d}{\delta} |M|. \quad (62)$$

Both large- m cases hold with

$$\rho_{\text{large}} = \frac{\delta}{d} = \frac{\theta^d}{2dK^d}. \quad (63)$$

Together with the small- m bound, $\rho = \theta^d/(2dK^d B_d^d)$ works for every m . Since $d/\gamma_d = 3d6^{d-2}$, equation (58) shows that the smaller explicit value

$$\rho = \frac{1}{2dK^d} \left(\frac{\varepsilon}{(C_0 d K)^{10d}} \right)^{3d6^{d-2}}$$

also works.

Undoing the global diagonal normalization preserves support and active-line sets. The theorem follows. □

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